Exponential Growth and Decay

Section 3.4

A. Population Growth

$$P(t) = P(0) \cdot e^{kt}$$

where

$P(t)$ = Population after t years

$P(0)$ = Initial Population

K = Growth constant

t = Time

Examples

1. **(video)** Solve $P(t) = P_0 \cdot e^{kt}$ for t.

2. **(video)** Solve the equation $9{,}000 = 1{,}500e^{kt}$ for t.

3. A bacteria culture initially contains 600 cells and grows at a rate proportional to its size. After 5 hours the population has increased to 620.

   a. Find an expression for the number of bacteria after t hours.

b. Find the number of bacteria after 7 hours.

c. Find the rate of growth after 7 hours. (Remember: Rate = Derivative)

d. When will the population reach 4,000?

B. Half-Life

$$P(t) = P(0) \cdot e^{kt}$$

where

$P(t)$ = Quantity after t years

$P(0)$ = Initial Quantity

K = Decay constant

t = Time

4. The half-life of cesium-137 is 30 years. Suppose we have a 900-mg sample.

a. Find the mass that remains after t years. (Find an expression for the mass that remains after t years.)

b. How much of the sample remains after 150 years?

c. After how long will only 4 mg remain?

d. Find the rate of decay after 20 years.

C. Newton's Law of Cooling

	Alternatively
$T(t)=(T_0-T_s)\cdot e^{kt}+T_s$	$T(t)=C\cdot e^{kt}+T_s$
where	where
$T(t)$ = Temperature after time t	$T(t)$ = Temperature after time t
T_s = Temperature of surrounding area	T_s = Temperature of surrounding area
T_0 = Initial temperature of object	C = Initial temperature − surrounding temperature
K = Growth constant	K = Growth constant
t = Time	t = Time

5. A roast turkey is taken from an oven when its temperature has reached 175 Fahrenheit and is placed on a table in a room where the temperature is 65 Fahrenheit.

a. If the temperature of the turkey is 155 Fahrenheit after half an hour, what is its temperature after 45 minutes?

b. When will the turkey have cooled to 110 Fahrenheit?

D. Interest

Compound Interest	Continuous Interest
$A = P\left(1+\frac{r}{n}\right)^{nt}$	$A = P \cdot e^{rt}$
where	where
$A =$ Future Value	$A =$ Future Value
$P =$ Initial Value	$P =$ Initial Value
$r =$ Interest rate	$r =$ Interest rate
$n =$ Number of times per year compounded	$t =$ Time in years
$t =$ Time in years	

6. **(video)** Find the accumulated value of an investment of \$120,000 at an interest rate of 4.5% if it is compounded monthly for 5 years.

7. **(video)** Find the accumulated value of an investment of \$120,000 at an interest rate of 4.5% if it is compounded continuously for 5 years.

8. If 8,000 dollars is invested at 9% interest, find the value of the investment at the end of 5 years if interest is compounded

a. Annually

b. Quarterly

c. Monthly

d. Continuously